United States Patent Application Publication	20250284148
Kind Code	A1
Publication Date	September 11, 2025
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WEARABLE DEVICE AND EARBUD

Abstract

A wearable device includes a main body, a detection module and a controller. The detection module is located on the main body and includes an upper casing, an opposite lower casing and a proximity sensor. The lower casing has an elastic material. The proximity sensor is located on the upper casing to measure a distance from the proximity sensor to the elastic material. The controller is electrically connected to the detection module. The elastic material has a contact surface for the user to contact when the wearable device is worn on the user, thereby switching between an active mode and a standby mode of the wearable device. In addition, the present disclosure further includes an earbud.

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Family ID: **73751356**

Appl. No.: **19/215370**

Filed: **May 22, 2025**

Foreign Application Priority Data

TW	109122117	Jun. 30, 2020

Related U.S. Application Data

parent US division 17195662 20210309 parent-grant-document US 12332506 child US 19215370

Publication Classification

Int. Cl.: **G02C11/00** (20060101); **H04R1/10** (20060101)

U.S. Cl.:

CPC **G02C11/10** (20130101); **H04R1/1041** (20130101); **H04R1/1058** (20130101); **H04R1/1091** (20130101);

Background/Summary

RELATED APPLICATIONS [0001] The present application is a Divisional Application of the U.S. application Ser. No. 17/195,662, filed Mar. 9, 2021, which claims priority to Taiwan Application Serial Number 109122117, filed Jun. 30, 2020, all of which are herein incorporated by reference in their entireties.

BACKGROUND
Field of Disclosure
[0002] The present disclosure relates to wearable devices and earbuds.
Description of Related Art
[0003] With the advancement of technology, various wearable devices that can be worn on users have become more common, such as smart watches, smart glasses, and so on. These wearable devices must be able to sense whether the user is using the wearable device in order to switch between different operating modes in real time, or intelligently enter standby mode to reduce power consumption.
[0004] Therefore, one of the problems that those skilled in the related art want to solve is how to provide a wearable device that can sense whether the user is wearing it in real time with low power consumption and is not easily false triggered.
SUMMARY
[0005] An aspect of the present disclosure is related to a wearable device.
[0006] According to one embodiment of the present disclosure, a wearable device includes a main body, a detection module and a controller. The detection module is located on the main body. The detection module includes a proximity sensor, an upper casing and an opposite lower casing. The lower casing has an elastic material. The proximity sensor is located on the upper casing to measure a distance from the proximity sensor to the elastic material. The controller is electrically connected to the detection module. The elastic material has a contact surface for a user to contact when the wearable device is worn on the user, thereby switching between an active mode and a standby mode of the wearable device.
[0007] In one or more embodiments, an elastic coefficient of the elastic material is different from an elastic coefficient of other portions of the lower casing beyond the elastic material.
[0008] In one or more embodiments, the elastic material is squeezed to approach the proximity sensor when the wearable device is worn on the user, so that a distance between the elastic material and the proximity sensor is reduced.
[0009] In one or more embodiments, the detection module further includes a light reflection layer. The light reflection layer is located on the elastic material. The proximity sensor is aligned with the light reflection layer.
[0010] In some embodiments, the proximity sensor has a light emitter and a light detector, the light emitter and the light detector faces the elastic material.
[0011] In some embodiments, the detection module further includes a timing unit configured to measure a plurality of measurement times spent by a plurality of measurement histories by a time interval and correspondingly convert the measurement times to a plurality of measurement data. A single one of the measurement histories is defined as a light emitted from the light emitter and reflected by the elastic material to the light detector.
[0012] In some embodiments, the detection module is configured to determine a first threshold and a second threshold greater than the first threshold to define active mode and the standby mode of

the wearable device. The wearable device switches to the active mode to operate when at least one of the measurement data is greater than the second threshold. The wearable device switches to the standby mode to operate when at least one of the measurement data is less than the first threshold during the active mode.

[0013] In some embodiments, the controller is configured to determine a first continuous time and a second continuous time. The wearable device switches to the active mode when any of the measurement data are greater than the second threshold during the first continuous time. The wearable device switches to the standby mode when any of the measurement data are less than the first threshold during the second continuous time.

[0014] In some embodiments, the time interval is configured to be selected from a first measurement interval and a second measurement interval less than the first measurement interval by the controller. The controller is configured to select the first measurement interval as the time interval when the wearable device is during the standby mode. The controller is configured to select the second measurement interval as the time interval when at least one of the measurement data is greater than the second threshold during the standby mode until the wearable device switches to the standby mode from the active mode.

[0015] In one or more embodiments, the main body further includes a glasses frame which is worn on the user, and the glasses frame has two temples arranged on both sides of a face of the user and two temple tips respectively extended from the two temples to abut against ears of the user. The detection module is located on one of the two temples and the two temple tips.

[0016] In some embodiments, an overall thickness of the detection module is less than or equal to a thickness of one of the two temples and the two temple tips.

[0017] In some embodiments, the controller is disposed in the two temples and connected to the detection module through a flexible circuit board.

[0018] Another aspect of the present disclosure is related to an earbud.

[0019] According to one embodiment of the present disclosure, an earbud includes a casing, a speaker, a detection module and a controller. The speaker is located in the casing to output audio signal. The detection module is located on the casing. The detection module includes an elastic material and a proximity sensor. The elastic material is located on the casing. The proximity sensor located on a support portion in the casing to measure a distance from the proximity sensor to the elastic material. The proximity sensor is aligned with the elastic material. The controller is located in the casing and electrically connected to the detection module. The elastic material has a contact surface for a user to contact when the earbud is worn on the user, thereby switching between an active mode and a standby mode of the earbud.

[0020] In one or more embodiments, the casing includes a body portion and an insertion portion connected to each other. The detection module is located on the casing and disposed at a junction between the body portion and the insertion portion.

[0021] In one or more embodiments, the elastic material is squeezed to approach the proximity sensor when the earbud is worn on the user, so that a distance between the elastic material and the proximity sensor is reduced.

[0022] In one or more embodiments, the proximity sensor has a light emitter aligned with the elastic material and a light detector.

[0023] In some embodiments, the detection module further includes a timing unit configured to measure a plurality of measurement times spent by a plurality of measurement histories by a time interval and correspondingly convert the measurement times to a plurality of measurement data. A single one of the measurement histories is defined as a light emitted from the light emitter and reflected by the elastic material to the light detector.

[0024] In some embodiments, the detection module is configured to determine a first threshold and a second threshold greater than the first threshold to define the active mode and the standby mode of the earbud. The earbud switches to the active mode to operate when at least one of the measurement data is greater than the second threshold. The earbud switches to the standby mode to operate when at least one of the measurement data is less than the first threshold during the active mode.

[0025] In summary, novel wearable device and earbud are provided in the present disclosure. The novel wearable device and earbud are switched between an active mode and a standby mode through a novel detection module, and power consumption can be reduced. By setting the first threshold and the second threshold, false triggering of the earbud and earbud due to walking or other vibrations can be avoided.

[0026] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The advantages of the present disclosure are to be understood by the following exemplary embodiments and with reference to the attached drawings. The illustrations of the drawings are merely exemplary embodiments and are not to be considered as limiting the scope of the present disclosure.

[0028] FIG. **1**A illustrates a schematic cross-section view of a detection module according to one embodiment of the present disclosure;

[0029] FIG. **1**B illustrates a schematic view that the detection module of FIG. **1**A detects a force signal;

[0030] FIG. **2** illustrates a block diagram that a circuit board of a detection module is connected to a micro controller board according to one embodiment of the present disclosure;

[0031] FIG. **3** illustrates a flow chart of a controlling method used for a wearable device according to one embodiment of the present disclosure;

[0032] FIGS. **4** to **6** respectively illustrate the relationship between different infrared reflection data and time corresponding to different force signals detected by a detection module of the present disclosure;

[0033] FIG. **7** illustrates a schematic view of a wearable device according to one embodiment of the present disclosure; and

[0034] FIG. **8** illustrates a schematic view of an earbud according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

[0035] The following embodiments are disclosed with accompanying diagrams for detailed description. For illustration clarity, many details of practice are explained in the following descriptions. However, it should be understood that these details of practice do not intend to limit the present invention. That is, these details of practice are not necessary in parts of embodiments of the present invention. Furthermore, for simplifying the drawings, some of the conventional structures and elements are shown with schematic illustrations. Also, the same labels may be regarded as the corresponding components in the different drawings unless otherwise indicated. The drawings are drawn to clearly illustrate the connection between the various components in the embodiments, and are not intended to depict the actual sizes of the components.

[0036] In addition, terms used in the specification and the claims generally have the usual meaning as each terms are used in the field, in the context of the disclosure and in the context of the particular content unless particularly specified. Some terms used to describe the disclosure are to be discussed below or elsewhere in the specification to provide additional guidance related to the description of the disclosure to specialists in the art.

[0037] The phrases “first,” “second,” etc., are solely used to separate the descriptions of elements or operations with the same technical terms, and are not intended to convey a meaning of order or to limit the disclosure.

[0038] Additionally, the phrases “comprising,” “includes,” “provided,” and the like, are all open-ended terms, i.e., meaning including but not limited to.

[0039] Further, as used herein, “a” and “the” can generally refer to one or more unless the context particularly specifies otherwise. It will be further understood that the phrases “comprising,” “includes,” “provided,” and the like used herein indicate the stated characterization, region, integer, step, operation, element and/or component, and does not exclude additional one or more other characterizations, regions, integers, steps, operations, elements, components and/or groups thereof.

[0040] The wearable device of the present disclosure includes a main body and a detection module. The main body includes objects that can be worn by the user, including wristbands, watches, glasses, earbuds, etc., but not limited to the present disclosure. By arranging a detection module of the present disclosure on the wearable device to be worn by users, the detection module can sense whether the user wears wearable device in real time, thereby switching the wearable device to an active mode to enable the wearable device performing the main function.

[0041] FIG. **1**A illustrates a schematic cross-section view of a detection module **110** according to one embodiment of the present disclosure. As shown in FIG. **1**A, in this embodiment, the detection module **110** includes an upper casing **115**, a lower casing **118** opposite to the upper casing **115**, an elastic material **119** located on the lower casing **118**, a circuit board **130** located on the upper casing **115** and aligned with the elastic material **119** and a proximity sensor **140** located on the circuit board **130**. The proximity sensor **140** faces the elastic material **119**. The opposite upper casing **115** and lower casing **118** define a chamber **120**. The circuit board **130** and the proximity sensor **140** are located in the chamber **120**.

[0042] The circuit board **130** can be a flexible circuit board. The circuit board **130** can be provided with elements having functions beyond the proximity sensor **140**. Additionally, the detection module **110** of the present disclosure can further include the controller **165** electrically connected to the proximity sensor **140** (details shown in following FIG. **2**).

[0043] The proximity sensor **140** is configured to measure a distance between the proximity sensor **140** and the elastic material **119**. In this embodiment, the proximity sensor **140** consists of an infrared emitter **145** and an infrared sensor **150**, and the infrared emitter **145** is aligned with the elastic material **119** to emit infrared light L to the elastic material **119**. The infrared light L is reflected to the infrared sensor **150** aligned with the elastic material **119** by the elastic material **119**.

[0044] As shown in FIG. **1**A, in this embodiment, a light reflection layer **125** is further located on the elastic material **119** to reflect the infrared light L.

[0045] The detection module of the present disclosure can include a light emitter and a light detector. In this embodiment, the infrared emitter **145** is used as the light emitter and the infrared sensor is used as the light detector, so that emitted light used for detection is infrared light L. However, this embodiment is not limited types of the light emitter and the light detector of the present disclosure. People in relevant fields can select light beyond the infrared light to detect changing of the distance by refection. In addition, this embodiment is not limited types of the proximity sensor **140**, and the present disclosure would include other elements that can realize the sensing of the distance between the elastic material **119** and the proximity sensor **140**.

[0046] In this embodiment, the light reflection layer **125** is a colored material, so as to effectively reflect infrared light. The light reflection layer **125** of colored material includes, for example, a white tape or a dark tape, but is not limited thereto. The colored tape can be provided on the elastic material **119** in the chamber **120** by means of attachment. As shown in FIG. **1**A, since the light reflection layer **125** is an adhesive tape, the light reflection layer **125** is close to the elastic material **119** and can be bent in response to changes of the elastic material **119**.

[0047] In some embodiments, the elastic material **119** can be made of material that facilitate reflection to directly reflect the infrared light L emitted by the infrared emitter **145** without disposing the light reflection layer **125**. For example, in some embodiments, the elastic material **119** with a dark color can be selected. In some embodiments, an elastic material **119** close to black is selected.

[0048] In some embodiments, if there is no interference from other light sources, the proximity sensor **140** (for example, including the infrared emitter **145** and the infrared sensor **150**) does not have to be disposed in the enclosed chamber **120**, but may be disposed in an open space. In other words, the upper casing **115** and the lower casing **118** do not necessarily form a closed chamber **120** if there is no interference from other light sources.

[0049] An elastic coefficient of the elastic material **119** is different from an elastic coefficient of other portions of the lower casing **118** beyond the elastic material **119**. In this embodiment, the elastic material **119** has an elastic coefficient less than the an elastic coefficient of other portions of the lower casing **118** beyond the elastic material **119**, so that the elastic material **119** is easier to be deformed by force than other portions of the lower casing **118** beyond the elastic material **119**. In some embodiments, material of the elastic material **119** is different from the material of the lower casing, so that the elastic coefficient of the elastic material **119** is different from the elastic coefficient of the lower casing **118** beyond the elastic material **119**. In some embodiments, the elastic material **119** and the lower casing **118** are the same material with different thicknesses and have different elastic coefficients. For details, please refer to following FIG. **1**B.

[0050] As shown in FIG. **1**A, in this embodiment, the elastic material **119** protrudes from a bottom of the lower casing **118**. The elastic material **119** has a contact surface **119**S exposed from the bottom of the lower casing **118**, so that the contact surface **119**S can be contacted with the user when the user wears a device with the detection module **110**. Then, the user can apply force to the elastic material **119** from outside the chamber **120**, so that the elastic material **119** is deformed and a distance between the elastic material **119** and the proximity sensor **140** is reduced. In some embodiments, a shape of the elastic material **119** on the lower casing **118** is a plane, and the elastic material **119** does not protrude from the bottom of the lower casing **118**, and the elastic material **119** is deformed and recessed into the chamber **120** after the user applies force, so that a distance between the elastic material **119** and the proximity sensor **140** is reduced. It should be noted that other configurations that reduce the distance due to wearing are also within the scope of the present disclosure. In some embodiments, the elastic material **119** and the lower casing **118** are integrally formed. In some embodiments, an opening can be provided in the lower casing **118**, and the elastic material **119** can be placed in the opening of the lower casing **118**.

[0051] Reference is made in FIG. **1**B to further describe the operations of the detection module **110**. FIG. **1**B illustrates a schematic view that the detection module **110** of FIG. **1**A detects a force signal.

[0052] As shown in FIG. **1**B, for the detection module **110** of the present disclosure, since the elastic material **119** protrudes from the bottom of the lower casing **118**, once the user touches the detection module **110** from the lower casing **118**, the user would inevitably touch the contact surface **119**S of the elastic material **119** first and apply force F on the contact surface **119**S, so that the elastic material **119** is deformed by the force F. At the same time, since the lower casing **118** has different elastic coefficient with respect to the elastic material **119**, the lower casing **118** is less deformed or even not deformed, as shown in FIG. **1**B.

[0053] Once the elastic material **119** is deformed by the force F, the distance between the proximity sensor **140** and the elastic material **119** changes. Correspondingly, the path taken by the infrared light L′ emitted by the infrared emitter **145** and reflected to the infrared sensor **150** in FIG. **1**B takes less time than the path taken by the infrared L in FIG. **1**A. Accordingly, it can be determined whether the user applies force F to the detection module **110** of a device of the present disclosure. When the user wears the wearable device with the detection module **110**, the elastic material **119** is squeezed and close to the proximity sensor **140**, so that the distance between the elastic material

and the proximity sensor **140** is reduced. Therefore, it can be determined whether the user wears the wearable device of the present disclosure.
[0054] In some embodiments, the upper casing **115** and the lower casing **118** can include a mixed material of polycarbonate (PC) and acrylonitrile-butadiene-styrene (ABS), and the mixed material can have a great hardness and elastic coefficient. The elastic material **119** can be selected from the group consisting of Thermoplastic elastomer (TPE), Thermoplastic polyurethane (TPU), Thermoplastic rubber (TPR) or other materials capable of greater elastic deformation than the lower casing **118**.
[0055] Please return to FIG. **1**A. In this embodiment, each of the upper casing **115** and the lower casing **118** can have a thickness of about 1 mm, so that the overall thickness of the detection module **110** can be greater than 2 mm. A distance between the upper casing **115** and the lower casing **118** corresponds to the distance between the proximity sensor **140** and the elastic material **119**, so that the accuracy of the detection module **110** are related the distance between the upper casing **115** and the lower casing **118**. A greater distance between the upper casing **115** and the lower casing **118** should better detect the deformation of the elastic material **119**, but would also increase the overall thickness H of the detection module **110**. Therefore, the overall thickness H of the detecting module **110** can be designed based on the type of the main body to be installed.
[0056] On the other hand, as shown in FIG. **1**A, in this embodiment, the thickness of the elastic material **119** is less than the thickness of the lower casing **118**. For example, the thickness of the lower casing **118** is 1 mm, and the thickness of the elastic material **119** may be in a range from 0.6 mm to 1.0 mm. In addition, for the case where the main body of the wearable device is a glasses frame, the width W.sub.F of the elastic material **119** can be configured to about 3 mm, but it is not limited to the present disclosure. The width W.sub.F of the elastic material **119** of the lower casing **118** is about 3 mm, which can surely contact the ears of the user when worn. Correspondingly, in some embodiments, the overall width of the detection module is greater than 3 mm but not limited to the present disclosure.
[0057] FIG. **2** illustrates a block diagram that a circuit board **130** of the detection module **110** is connected to a micro controller board **160** according to one embodiment of the present disclosure. As shown in FIG. **1**, the circuit board **130** is located on the upper casing **115** of the detection module **110**. In FIG. **2**, the circuit board **130** is substantially connected to the micro control board **160** to electrically connect to the controller **165** on the micro control board **160**.
[0058] As shown in FIG. **2**, in this embodiment, the circuit board **130** has a flash memory **134** and a timing unit **132** connected to each other. The timing unit **132** is connected to the infrared emitter **145** and the infrared sensor **150** respectively. The timing unit **132** can be used to record the time taken from the infrared emitter **145** emitting infrared light (such as infrared light L or L′) to the infrared sensor **150** receiving infrared light, and the time would be recorded in the flash memory **134**. In some embodiments, the flash memory **134** and the timing unit **132** can be substantially integrated into a microcontroller unit.
[0059] Once the elastic material **119** is applied with force and deformed, the distance between the elastic material **119** and the proximity sensor **140** would also be changed. For example, the path of the infrared light L in FIG. **1**A is different from the path of the infrared light L′. The changed distance between the elastic material **119** and the proximity sensor **140** would cause the record time recorded by the timing unit **132** being different, so that the change in the distance between the elastic material **119** and the proximity sensor **140** can be reflected in the time recorded by the timing unit **132**.
[0060] Therefore, the timing unit **132** can be used to measure a plurality of measurement times spent by a plurality of measurement histories by a time interval and correspondingly convert the measurement times into measurement data. In this embodiment, a single one of the measurement histories is defined as an infrared light emitted from the infrared emitter **145** and reflected to the infrared sensor **150** through the elastic material **119**. The measurement data obtained by the timing

unit **132** is a countdown of infrared reflection data. When the infrared emitter **145** emits infrared light, the timing unit **132** starts counting down from a preset value until the infrared sensor **150** receives the reflected infrared light and stops the countdown to obtain corresponding infrared reflection data.

[0061] Each measurement history can occur every time interval. For example, if the time interval is configured to 1 second, during the first (1.sup.st) second, a first infrared reflection data is defined as an infrared light emitted by the infrared emitter **145** is received by the infrared sensor **150** and recorded. After a one-second interval and during the second (2.sup.nd) second, a second infrared reflection data is defined as an infrared light emitted by the infrared emitter **145** is received by the infrared sensor **150** and recorded.

[0062] In this embodiment, an infrared reflection data is defined by a countdown operation. Therefore, an infrared (IR) reflection data is a less value if the distance between the proximity sensor **140** and the elastic material **119** is greater since the path passed by the infrared light L is greater and the timing unit **132** would count down more time, as shown in FIG. **1**A. Similarly, in FIG. **1**B, the elastic material **119** is deformed by the force F, so that the distance between the proximity sensor **140** and the elastic material **119** is reduced, the path of the infrared light L′ is less, and the timing unit **132** would count down less, thereby obtaining a greater infrared reflection data.

[0063] In this embodiment, the micro control board **160** is provided with a controller **165**, a charging module **167**, a hint light **169**, a flash memory **171**, an antenna module **173**, and a control interface **175**. The controller **165** is electrically connected to the detection module **110** through the detection module port to receive measurement data (i.e. Infrared reflection data) provided by the timing unit **132** to sense whether the user is wearing the wearable device of the present disclosure, for example. In this embodiment, the controller **165** is a Bluetooth chip but not limited to the present disclosure.

[0064] The controller **165** is connected to the charging module **167** through a charging port. The controller **165** is connected to the charging module **167** through a charging port to obtain power through an external power source or the like. The controller **165** is connected to the hint light **169** through the hint light port. The hint light **169** is used to show the current state of the wearable device of the present disclosure for the user, such as charging, standby mode, active mode, and so on. The controller **165** is connected to the flash memory **171** through a memory port to receive the time change signal generated by one or more force from the detection module **110**. The controller **165** is connected to the antenna module **173** through the antenna port to receive other external signals. For example, the antenna module **173** includes a Bluetooth antenna, such as a 2.4 G Bluetooth antenna. The controller **165** is also connected to the control interface **175** through the control interface port. The control interface **175** can be provided with a plurality of physical buttons or touch control surfaces, and the user can actively control the wearable device of the present disclosure through the control interface **175**.

[0065] To further illustrate that how the controller **165** receive the time change signal of the detection module **110** and controls the wearable device of the present disclosure, please refer to FIG. **3**. FIG. **3** illustrates a flow chart of a controlling method **300** used for a wearable device according to one embodiment of the present disclosure. Also refer to FIGS. **4** to **6**. FIGS. **4** to **6** respectively illustrate the relationship between different infrared reflection data and time corresponding to different force signals detected by a detection module **110** of the present disclosure.

[0066] Before accessing the controlling method **300** specifically controls the wearable device of the present disclosure, the user can preset a plurality of controlling parameters in the detection module **110** and the controller **165**. Subsequently, the controlling method **300** can switch the wearable device between the active mode and the standby mode through the controller **165** according to these controlling parameters. Specifically, in this embodiment, these controlling parameters include: a first measurement interval TM.sub.1 and a second measurement interval TM.sub.2, a first threshold value TH.sub.1 and a second threshold value TH.sub.2, a first continuous time T.sub.1 and a second continuous time T.sub.2.

[0067] As mentioned above, the timing unit **132** of the wearable device of the present disclosure is used for timing a plurality of measurement histories at a time interval to obtain the measurement data. In this embodiment, a single measurement history is defined as an infrared light emitted from the infrared emitter **145** and reflected until the infrared light is received by the infrared sensor **150**, and the measurement data is the infrared reflection data calculated by counting down the preset value.

[0068] Therefore, in this embodiment, the controller **165** is configured to select the time interval from the first measurement interval TM.sub.1 and the second measurement interval TM.sub.2. The second measurement interval TM.sub.2 is less than the first measurement interval TM.sub.1 and corresponds to denser measurement. As such, the controller **165** has to switch between the first measurement interval TM.sub.1 and the second measurement interval TM.sub.2 in time to reduce power consumption.

[0069] For the infrared reflection data measured by the detection module **110**, a first threshold TH.sub.1 and a second threshold TH.sub.2 greater than the first threshold TH.sub.1 can be configured in the detection module **110** and the controller **165**. As mentioned above, since the infrared reflection data is obtained by counting down the preset value, the closer the proximity sensor **140** and the elastic material **119** in the detection module **110** are, the greater the corresponding infrared reflection data value. Therefore, the greater infrared reflection data corresponds to the user wearing a wearable device, and the less infrared reflection data corresponds to the user not wearing the wearable device. It can be regarded that the user has worn the wearable device if the infrared reflection data is greater than the second threshold TH.sub.2, and it can also be regarded that the user does not wear the wearable device if the infrared reflection data is less than the first threshold TH.sub.1. The range between the first threshold TH.sub.1 and the second threshold TH.sub.2 can be used to avoid false triggering from the user.

[0070] To avoid false triggering, a first continuous time T.sub.1 can also be configured by the controller **165**, so that it is recognized as the corresponding user is wearing the wearable device if the infrared reflection data are greater than both of the first threshold TH.sub.1 and the second threshold TH.sub.2 in the first continuous time T.sub.1. Similarly, a second continuous time T.sub.2 can also be configured by the controller **165**, so that it is recognized as the corresponding user is not wearing the wearable device if the infrared reflection data is less than the first threshold TH.sub.1 in the second continuous time T.sub.2.

[0071] In this embodiment, the first measurement interval TM.sub.1 is set to 1 second, the second measurement interval TM.sub.2 is set to 0.1 seconds, the first threshold TH.sub.1 is set to 14040 (in counts), and the second threshold TH.sub.2 greater than the first threshold TH**1** is set to 14120 (in counts). The first continuous time T.sub.1 is set to 2 seconds, and the second continuous time T.sub.2 is set to 2 minutes. However, this embodiment is not limited to the present disclosure.

[0072] As shown in FIG. **3**, the controlling method **300** includes operations **310** to **350**. In operation **310**, starting the power supply of the wearable device of the present disclosure, wherein the starting includes starting the detection module **110** and the controller **165** on the micro control board **160** as described above.

[0073] Proceed to operation **315** and after starting, put the wearable device into a standby mode.

[0074] In the standby mode, the operation **320** is entered, and the detection module **110** uses the first measurement interval TM.sub.1 (1 second) as the time interval to continuously obtain infrared reflection data (measurement data). In other words, the controller **165** of the detection module **110** makes the detection module **110** measure data every 1 second to obtain the measurement data. Specifically, the infrared emitter **145** of the detection module **110** is made to emit infrared light at an interval of 1 second, and the infrared light are reflected by the elastic material **119** and received by the infrared sensor **150**.

[0075] During continuously obtaining infrared reflection data (measurement data) with the first measurement interval TM.sub.1 (1 second) as the time interval, enter the operation **325** to confirm whether the infrared reflection data (measurement data) is greater than the second threshold TH.sub.2. If not, return to the operation **315**, and the wearable device remains in the standby mode. If so, the operation **330** is entered, and the detection module **110** uses the second measurement interval TM.sub.2 (0.1 second) as the time interval to obtain the infrared reflection data (measurement data) more intensively. Regarding operation **320** to operation **330**, please refer to FIG. **4** for details.
[0076] As shown in FIG. **4**, the vertical axis is the infrared reflection data recorded by the timing unit **132** in the countdown of the preset value, and the unit of the vertical axis is count. The horizontal axis is the time, and the unit of the horizontal axis is second.
[0077] Proceed to operation **325** and if no, there is no infrared reflection data greater than the first threshold TH.sub.1 or the second threshold TH.sub.2 at the 0th second. Therefore, keep the wearable device in standby mode and still use a longer first measurement interval TM.sub.1 (1 second) as the time interval.
[0078] Proceed to operation **325** and if yes, the infrared reflection data is detected which is greater than the second threshold TH.sub.2 at the 1st second after the first measurement interval TM.sub.1 (1 second) has flied, so the time interval of the detection module **110** is switched to the second measurement interval TM.sub.2 (0.1 second) to obtain the infrared reflection data at different time points more intensively.
[0079] The controller **165** can be still activated in a low power consumption mode during the standby mode. Until the operation **330** is entered, the controller **165** is switched to operate in the normal power consumption mode. Therefore, the power consumption of the controller **165** can be reduced.
[0080] Reference is made in FIGS. **3** and **4**. In operation **335** and during the first continuous time T.sub.1 (2 seconds), the controller **165** continuously confirms whether the infrared reflection data are all greater than the second threshold TH.sub.2. If not, it is recognized as a false triggering of the user and returns to the operation **315** to keep the wearable device in the standby mode. If yes, enter to the operation **340**, and switch the wearable device to the active mode.
[0081] During the active mode, enter the operation **345** to confirm whether the infrared reflection data (measurement data) is greater than the second threshold TH.sub.2. If the infrared reflection data is always greater than the second threshold TH.sub.2, return to the operation **340**, and the wearable device maintains the active mode. If not, enter the operation **350** to confirm whether the infrared reflection data (measurement data) are all less than the first threshold TH.sub.1 in the second continuous time T.sub.2 (2 minutes). Please refer to FIGS. **5** and **6** corresponding to two different situations.
[0082] Reference is made in FIG. **5**. In the second continuous time T.sub.2 (2 minutes), although part of the infrared reflection data is less than the second threshold TH.sub.2, the infrared reflection data is still maintained to be greater than the first threshold TH.sub.1. Based on such situation, the controller **165** can recognize that the user may cause the infrared reflection data to be bounced due to movement, so it is determined that the wearable device is still being worn, and the active mode should be maintained. In some embodiments, some of the infrared reflection data in the second continuous time T.sub.2 is less than the first threshold TH.sub.1. However, the controller **165** still recognizes the user is wearing when not all of the infrared reflection data in the second continuous time T.sub.2 are less than the first threshold TH.sub.1. This case corresponds to the operation **350** being No, returning to the operation **340** to maintain the wearable device in the active mode.
[0083] On the contrary, reference is made in FIG. **6**. In the second continuous time T.sub.2 (2 minutes), if all the infrared reflection data are less than the first threshold TH.sub.1, the controller **165** recognizes that the user is not wearing the wearable device. This case corresponds to the operation **350** being Yes, returning to the operation **315** to enter the standby mode again. The

controller **165** is switched to the standby mode, which is the low power consumption mode, and the detection module **110** uses the first measurement interval TM.sub.1 as the time interval again to obtain infrared reflection data (operation **320**).

[0084] Therefore, through the controlling method **300**, the wearable device of the present disclosure can recognize whether the user is wearing it through the detection module **110** and the controller **165**, thereby automatically switching between the active mode and the standby mode for reducing power consumption.

[0085] To further illustrate a specific example of a wearable device of the present disclosure, please refer to FIG. **7**. FIG. **7** illustrates a schematic view of a wearable device **100** according to one embodiment of the present disclosure.

[0086] As shown in FIG. **7**, in this embodiment, the main body provided with the detection module **110** or the detection module **110′** is a glasses frame **180**. The glasses frame **180** is provided for the user to wear.

[0087] In FIG. **7**, the glasses frame **180** has two temples **183** arranged on both sides of the face of the user, and two temple tips **186** respectively extending from the two temples **183** to abut against the ears of the user. The detection module **110** can be arranged on one of the two temples **183** and the two temple tips **186**, and the contact surface **119**S of the elastic material **119** (please refer to FIG. **1**A) is used to contact the skin of the user. It should be noted that FIG. **7** only schematically illustrates the location in which the detection module **110** or the detection module **110′** can be located. For the purpose of simple description, the contact surface **119**S of the elastic material **119** of the detection module **110** or the detection module **110′** is not shown in FIG. **7**. FIG. **7** only schematically illustrates the location of the detection module **110** or the detection module **110′**, other locations that can be in contact with the user's skin are also included in this disclosure.

[0088] In this embodiment, the glasses frame **180** can be provided with only one of the detection module **110** and the detection module **110′**, and the micro control board **160** is provided in the temple **183**, so that the controller **165** on the micro control board **160** can be electrically connected to the detection module **110**. In other words, the controller **165** can be arranged on the micro control board **160** in the temple **183** and electrically connected to the detection module **110** through a flexible circuit board. As shown in FIG. **2**, other components of the micro control board **160** (including the hint light **169** and the buttons of the control interface **175**, etc.) can also be exposed from the temple **183** for confirmation and operation for the user.

[0089] The detection module **110** is located at the junction of the temple **183** and the temple tip **186**. When the user wears the glasses frame **180**, the ears of the user would abut the junction between the temple **183** and the temple tip **186**, thereby contacting the detection module **110**. To contact with the ear of the user, the contact surface **119**S of the elastic material **119** of the detection module **110** should be facing down, so that the contact surface **119**S is exposed from the bottom of the junction and contacts the ear of the use, and the elastic material **119** is deformed by force to have a greater infrared reflection data measured by the detection module **110**.

[0090] The detection module **110′** is located in the temple **183**. When the user wears the glasses frame **180**, the face of the user would contact the temple **183**. To contact with the skin of the user, the contact surface **119**S of the elastic material **119** of the detection module **110′** should face the face of the user, so that the contact surface **119**S can contact the skin of the user.

[0091] In order to dispose the detection module **110** or the detection module **110′** in the glasses frame **180** as shown in FIG. **7**, the overall thickness H of the detection module **110** or the detection module **110′** should be less than a thickness H.sub.1 of the temple **183** or the thickness H.sub.2 of the temple tip **186**.

[0092] Therefore, the wearable device **100** can be used as smart glasses or wireless audio glasses, and the controlling method **300** can be realized by the detection module **110** or the detection module **110′** to confirm whether the user is wearing the wearable device **110**, so as to automatically switch the wearable device **110** between the active mode and the standby mode. In the active mode,

the wearable device **100** can perform other main functions of the glasses, such as turning on Bluetooth or playing audio, etc. In the standby mode, the wearable device **100** can, for example, turn off the Bluetooth, stop playing audio, etc., to reduce power consumption.
[0093] FIG. **8** illustrates a schematic view of an earbud **200** according to one embodiment of the present disclosure. The earbud **200** is a type of wearable device of the present disclosure and is provided with a detection module **110″**. In some embodiments, the detection module **110″** can also be similar to the detection module **110** shown in FIGS. **1**A to **2** so as to be directly integrated in the earbud **200**. It should be noted that FIG. **8** is only a schematic drawing of the location in which the detection module **110″** is located. Other locations that can be in contact with the user are also included in the present disclosure.
[0094] As shown in FIG. **8**, the earbud **200** includes a casing **230**, a speaker **210**, and a detection module **110″**. The casing **230** includes a body portion **235** and an insertion portion **240** connected to each other. The speaker **210** is installed in the insertion portion **240**, and the controller **250** of the detection module **110″** and the battery **220** of the earbud **200** are both disposed in the body portion **235**.
[0095] In FIG. **8**, the detection module **110″** is located on the casing **230** and disposed at the junction of the body portion **235** and the insertion portion **240**. In this embodiment, the detection module **110″** includes an elastic material **119** and a proximity sensor **140** aligned with the elastic material **119**. In order to contact the user, the elastic material **119** of the detection module **110″** is located on the casing **230**. A contact surface **119**S (as shown in FIG. **1**A) of the elastic material **119** is exposed outside the casing **230** and can be in contact with the user. The proximity sensor **140** of the detection module **110″** can be located on a support portion in the casing **230**. In some embodiments, the support portion is a hard material that can be loaded with the circuit board **130** and the proximity sensor **140**. The function of the support portion is similar to the upper case **115** shown in FIG. **1**A and used to support the proximity sensor **140**. In some embodiments, the support portion can also be a part extending from the casing **230**. For the purpose of simple description, the support portion and the contact surface **119**S, etc. are not shown in FIG. **8** but only schematically indicate the position of the detection module **110″** on FIG. **8**. In summary, in this embodiment, the support portion of the detection module **110″** in the earbud **200** can be a hard material with the circuit board **130** and the proximity sensor **140** in the casing **230**. The elastic material **119** can be integrally formed on the casing **230** of the earbud **200**, and the proximity sensor **140** faces the elastic material **119**.
[0096] Therefore, when the user wears the earbud **200**, the elastic material **119** is squeezed and approaches the proximity sensor **140**, so that the distance between the elastic material **119** and the proximity sensor **140** is shortened. As mentioned above, the proximity sensor **140** can include a light emitter (such as the infrared emitter **145**) and a light detector (such as the infrared sensor **150**) faces the elastic material **119**. As such, the controlling method **300** can be used to switch the earbud **200** between the active mode and the standby mode.
[0097] In this embodiment, the earbud **200** includes a wireless earphone, a true wireless stereo (TWS), a headset, etc., but not limited to the present disclosure. When switched to the active mode, the earbud **200** can perform main functions such as turning on the Bluetooth or playing audio, etc., and when switched to the standby mode, the earbud **200** can turn off the Bluetooth or stop playing audio, etc. to reduce power consumption.
[0098] In some embodiments, a wearable device or an earbud has two opposite surfaces on which the position of the proximity sensor **140** and the elastic material **119** can be respectively arranged is included within the scope of the present disclosure.
[0099] In summary, the present disclosure provides wearable devices, such as glasses and earbud (e.g. wireless earphone) with an improved detection module. By integrating the proximity sensor and the elastic material deformed by force, the improved detection module can easily know whether the user is wearing the wearable device. In addition, through the designed controlling

method, the active mode and standby mode of the wearable device can be switched instantly, effectively reducing power consumption. By setting the first and second thresholds, it is also possible to effectively avoid false triggering of the wearable device by the user.
[0100] Although the present invention has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims.

Claims

1. An earbud, comprising: a casing; a speaker located in the casing to output audio signal; a detection module located on the casing and comprising: an elastic material located on the casing; and a proximity sensor located on a support portion in the casing to measure a distance from the proximity sensor to the elastic material, wherein the proximity sensor is aligned with the elastic material; and a controller located in the casing and electrically connected to the detection module, wherein the elastic material has a contact surface for a user to contact when the earbud is worn on the user, thereby switching between an active mode and a standby mode of the earbud.
2. The earbud of claim 1, wherein the casing comprises a body portion and an insertion portion connected to each other, and the detection module is located on the casing and disposed at a junction between the body portion and the insertion portion.
3. The earbud of claim 1, wherein the elastic material is squeezed to approach the proximity sensor when the user wears the earbud, so that a distance between the elastic material and the proximity sensor is reduced.
4. The earbud of claim 1, wherein the proximity sensor has a light emitter and a light detector, the light emitter and the light detector faces the elastic material.
5. The earbud of claim 4, wherein the detection module further comprises a timing unit configured to measure a plurality of measurement times spent by a plurality of measurement histories by a time interval and correspondingly convert the measurement times to a plurality of measurement data, wherein a single one of the measurement histories is defined as a light emitted from the light emitter and reflected by the elastic material to the light detector.
6. The earbud of claim 5, wherein the detection module is configured to determine a first threshold and a second threshold greater than the first threshold to define the active mode and the standby mode of the earbud, wherein the earbud switches to the active mode to operate when at least one of the measurement data is greater than the second threshold, the earbud switches to the standby mode to operate when at least one of the measurement data is less than the first threshold during the active mode.
7. The earbud of claim 6, wherein the controller is configured to determine a first continuous time and a second continuous time, the earbud switches to the active mode when any of the measurement data are greater than the second threshold during the first continuous time, the earbud switches to the standby mode when any of the measurement data are less than the first threshold during the second continuous time.
8. The earbud of claim 6, wherein: the time interval is configured to be selected from a first measurement interval and a second measurement interval less than the first measurement interval by the controller, the controller is configured to select the first measurement interval as the time interval when the earbud is during the standby mode, the controller is configured to select the second measurement interval as the time interval when at least one of the measurement data is greater than the second threshold during the standby mode until the earbud switches to the standby mode from the active mode.
9. The earbud of claim 1, wherein an elastic coefficient of the elastic material is different from an

elastic coefficient of other portions of the casing beyond the elastic material.

10. The earbud of claim 1, wherein the detection module further comprises a light reflection layer, the light reflection layer is disposed on the elastic material, and the proximity sensor is aligned with the light reflection layer.